

Francisco Garre-Frutos

POSTDOCTORAL RESEARCHER · EXPERIMENTAL PSYCHOLOGY

Granada, Spain

+34 686 533 398 | fgfrutos@gmail.com | [franfrutos.github.io](https://github.com/franfrutos) | [franfrutos](https://www.linkedin.com/company/franfrutos) | [Scholar](https://scholar.google.com/citations?user=frfrutos) | [ORCID](https://orcid.org/0000-0001-9550-4444) | [OSF](https://osf.io/frfrutos) | [Bluesky](https://bluesky.com/frfrutos)

My work lies at the intersection of learning and cognitive control, with a particular focus on how reward history shapes visual attention.

Education

PhD in Psychology

UNIVERSITY OF GRANADA

Granada, Spain

2021 – 2025

Thesis: *Awareness, Control, and Automaticity in Value-Modulated Attentional Capture.*

Supervisors: Juan Lupiáñez and Miguel A. Vadillo.

Distinction: International Doctorate; cum laude.

MSc in Cognitive and Behavioral Neuroscience

UNIVERSITY OF GRANADA

Granada, Spain

2020 – 2021

Thesis: *Inhibition of return and learned value.*

Supervisor: Juan Lupiáñez.

BSc in Psychology

UNIVERSITY OF MURCIA

Murcia, Spain

2016 – 2020

Distinction: Outstanding Undergraduate Award.

Research Experience

Postdoctoral Researcher

UNIVERSITY OF GRANADA

Granada, Spain

Since Dec. 2025

Postdoctoral appointment at the Mind, Brain, and Behavior Research Center (CIMCYC). Funded through the University of Granada bridge postdoctoral contract programme.

Predoctoral Researcher

UNIVERSITY OF GRANADA

Granada, Spain

Nov. 2021 – Dec. 2025

My thesis examined automaticity in learned attentional biases, focusing on how signals of value can capture attention automatically and how this effect relates to awareness and control. Funded by an FPU predoctoral fellowship from the Spanish Ministry of Science and Innovation.

Visiting Researcher

VRIJE UNIVERSITEIT AMSTERDAM

Amsterdam, the Netherlands

Sept. 2024 – Dec. 2024

Three-month research stay in Jan Theeuwes's lab at Vrije Universiteit Amsterdam, where we collaborated on a project that extended my dissertation work on automaticity in attentional capture by signals of value ([preprint](#)). Funded by a complementary FPU mobility grant from the Spanish Ministry of Science and Innovation.

Research Scholarship

UNIVERSITY OF GRANADA

Granada, Spain

Sept. 2020 – Jul. 2021

Funded by a research initiation scholarship from the University of Granada. Worked with José César-Perales on individual differences in learning and decision-making among people with problematic videogame use.

Research Scholarship

UNIVERSITY OF MURCIA

Murcia, Spain

Sept. 2019 – Jul. 2020

Funded by a collaboration grant from the Spanish Ministry of Education. Worked with Víctor Martínez-Pérez using EEG and tDCS to study how control-related brain mechanisms relate to inhibitory processes in attention.

Methods & Technical Expertise

Specialized Training

Tenth Summer School on Statistical Methods for Linguistics and Psychology (SMLP): Advanced Methods in Frequentist Statistics with Julia

Aug. 2026

University of Potsdam, Germany · 40 hours.

Barcelona Summer School on Advanced Modeling of Behavior (BAMB 2026)

Jul. 2026

Centre de Recerca Matemàtica (CRM), Barcelona · 92.5 hours.

Electroencephalography (EEG): Preprocessing and Introduction to Univariate and Multivariate Analyses

Mar. 2025

Escuela de Doctorado de Ciencias de la Salud, University of Granada · 21 hours.

Annual JAGS Workshop: Bayesian Modeling for Cognitive Science

Jul. 2024

Psychological Methods Group, University of Amsterdam · 140 hours.

Seventh Summer School on Statistical Methods for Linguistics and Psychology (SMLP): Advanced Bayesian Data Analysis

Sept. 2023

University of Potsdam, Germany · 40 hours.

CIMCYC Workshop on Computational Modelling of Behavioral Data

Jun. 2022

Mind, Brain, and Behavior Research Center (CIMCYC), University of Granada · 9 hours.

Publications

- Franco-Martínez, A., Rey-Sáez, R., Adrián-Ventura, J., ..., Garre-Frutos, F., ..., & Vadillo, M. A. (2026). *Replicating the unconscious working memory effect: A multisite Registered Report*. *Neuroscience of Consciousness*, 2026(1), niaf046.



- Martínez-López, P., Vázquez-Millán, A., Garre-Frutos, F., & Luque, D. (2026). *Assessing the validity evidence for habit measures based on time pressure*. *Behavior Research Methods*, 58, Article 7.



- Garre-Frutos, F., Lupiáñez, J., & Vadillo, M. A. (2025). *Value-modulated attentional capture depends on awareness*. *Psychonomic Bulletin & Review*, 32, 3025–3040.



- Garre-Frutos, F., Ariza, A., & González, F. (2025). *The effect of reward and punishment on the extinction of attentional capture elicited by value-related stimuli*. *Psychological Research*, 89, Article 89.



- González, F., Garre-Frutos, F., Hinojosa-Aguayo, I., & Hall, G. (2025). *Extinction of outcome-specific Pavlovian-to-instrumental transfer (PIT), instrumental outcome devaluation, and reward-related attentional capture are predicted by affect-driven impulsivity*. *Learning & Behavior*, 53, 335–354.



- Garre-Frutos, F., Vadillo, M. A., González, F., & Lupiáñez, J. (2024). *On the reliability of value-modulated attentional capture: An online replication and multiverse analysis*. *Behavior Research Methods*, 56(6), 5986–6003.



OUTREACH PUBLICATION

- Garre-Frutos, F. (2025). *Is wanting enough? Control and automaticity in learned attentional biases*. *Ciencia Cognitiva*, 19(1), 8–10.

Ongoing Work

- Franco-Martínez, A., Garre-Frutos, F., & Tomás, G. (2026). *Measuring Creativity Process and Product with a Digitized Invented Alien Creatures Task*. *PsyArXiv*.



- Garre-Frutos, F., Vadillo, M. A., Theeuwes, J., Van Moorselaar, D., & Lupiáñez, J. (2025). *The role of selective attention in value-modulated attentional capture*. PsyArXiv; manuscript resubmitted to *Psychonomic Bulletin & Review*.
   
- Garre-Frutos, F., López-Martínez, P., Rey-Sáez, R., Luque, D., Vadillo, M. A., & González, F. (2026). *Explicit knowledge modulates Pavlovian attentional biases*. PsyArXiv.
   
- Franco-Martínez, A., Ferrer-Mendieta, A., Peiro-Lanchares, C., Vadillo, M. A., & Garre-Frutos, F. (2025). *Choose your own PAS? Quantitative and Qualitative Validity Evidence for the Perceptual Awareness Scale*. PsyArXiv; manuscript resubmitted to *Behavior Research Methods*.
   
- Theeuwes, J., Manini, G., Garre-Frutos, F., Lupiáñez, J., & van Moorselaar, D. *The capture debate revisited: Proactive or reactive suppression. That is the question*. Manuscript submitted to *Visual Cognition*.
   
- Manini, G., Garre-Frutos, F., Botta, F., Martínez-Arévalo, E., & Lupiáñez, J. (2025). *Distraction vs. Interference: Handling fully irrelevant vs. potentially relevant distractors*. PsyArXiv.
   
- Manini, G., Garre-Frutos, F., Botta, F., Martínez-Arévalo, E., & Lupiáñez, J. (2025). *Facilitation by Irrelevant Distractors Under High Perceptual Load: Suppression or Controlled Attentional Capture?*. PsyArXiv.
   
- Garre-Frutos, F., & Lupiáñez, J. (2024). *Inhibition of return and learned value*. PsyArXiv; manuscript submitted to *Psicológica*.
   

Funded Projects

Understanding the Learning and Unlearning Dynamics of Myths about (Cyber)Violence Against Women MARÍA DE MAEZTU UNIT OF EXCELLENCE – CIMCYC · UNLEARN-CVAW Role: Co-PI with Rocío Vizcaíno-Cuenca. Scheme: Support Program for New Collaborative Projects for Early-Career Researchers.	EUR 2,000 Since 2026
Visiting Scholars Programme UNIVERSITY OF GRANADA VISITING SCHOLARS PROGRAMME · PPVS2025-01 Visiting scholar: Jan Theeuwes. Responsible researcher: María Ruz Cámara. Team: Juan Lupiáñez, David Richter, and Francisco Garre-Frutos.	EUR 12,000 2026 – 2027
Let's ROC: A generalization of Signal Detection Theory for the study of unconscious mental processes AGENCIA ESTATAL DE INVESTIGACIÓN (MCIN/AEI) · PID2023-150830NB-I00 Role: Research team. PI: Miguel A. Vadillo, Universidad Autónoma de Madrid.	EUR 150,000 Since 2024
Conceptual Analysis of Vigilance Decrement: Attentional Resources, Cognitive Control, and Mind-Wandering AGENCIA ESTATAL DE INVESTIGACIÓN (MCIN/AEI) · PID2023-148421NB-I00 Role: Research team. PI: Juan Lupiáñez, University of Granada.	EUR 177,250 Since 2024
Experimentally investigating the formation of new habits in humans AGENCIA ESTATAL DE INVESTIGACIÓN (MCIN/AEI) · PID2021-126767NB-I00 Role: Research team. PI: David Luque, University of Málaga.	EUR 109,868 2022 – 2025

Role: Research team.

PI: Felisa González, University of Granada.

Theoretical, behavioral, and functional analysis of executive and vigilance and its implementation in neural networks

EUR 157,300

AGENCIA ESTATAL DE INVESTIGACIÓN (MCIN/AEI) · PID2020-114790GB-I00

2021 – 2024

Role: Research team.

PI: Juan Lupiáñez, University of Granada.

Conference Presentations

ORAL PRESENTATIONS

- Garre-Frutos, F., López-Martínez, P., Rey-Sáez, R., Luque, D., Vadillo, M. A., & González, F. (2026). *Explicit knowledge modulates Pavlovian attentional biases*. APS Annual Convention, Barcelona, Spain, May 28–30.
- Martínez-López, P., Garre-Frutos, F., León, J. J., Vázquez-Millán, A., Molinero, S., Pérez-Velasco, N., Vadillo, M. A., & Luque, D. (2025). *Lingering habits: Testing the temporal stability of habits*. XXXV International Congress of the Spanish Society for Comparative Psychology, Donostia / San Sebastián, Spain, October 1–3.
- Garre-Frutos, F., Solana, P., López-Martínez, P., Lupiáñez, J., & Vadillo, M. A. (2025). *Reliability and validity of experimental measures of value-driven attention: A meta-analysis on individual differences studies*. XI Conference of the European Association of Methodology, Tenerife, Spain, July 23–25.
- Franco-Martínez, A., Ferrer-Mendieta, A., Peiró-Lanchares, C., Garre-Frutos, F., & Vadillo, M. A. (2025). *Choose your own PAS? Studying the validity of the Perceptual Awareness Scale*. XI Conference of the European Association of Methodology, Tenerife, Spain, July 23–25.
- Martínez-López, P., Garre-Frutos, F., Luque, D., & Vadillo, M. A. (2025). *A systematic review and an internal consistency analysis of behavioral habit measures*. XI Conference of the European Association of Methodology, Tenerife, Spain, July 23–25.
- Garre-Frutos, F., Luque, D., Martínez-López, P., Lupiáñez, J., & Vadillo, M. A. (2025). *Modelling Pavlovian attentional biases through reinforcement learning*. 14th Reunión Científica sobre Atención, Madrid, Spain, April 3–4.
- Rey-Sáez, R., Garre-Frutos, F., & Vadillo, M. A. (2024). *What psychometrics can offer to experimental psychology: The multi-level model as an heir of the factorial approach*. XVIII Conference on Methodology of Social and Health Sciences, Seville, Spain, September 3–6.
- Garre-Frutos, F., Vadillo, M. A., & Lupiáñez, J. (2024). *An online replication and multiverse reliability analysis of value-modulated attentional capture*. XIV SEPEX Conference, Almería, Spain, October 23–25.
- Martínez-López, P., Garre-Frutos, F., Vadillo, M. A., & Luque, D. (2024). *Evaluation of the permutation-based split-half reliability in experimental paradigms for habit induction in humans*. XIV SEPEX Conference, Almería, Spain, October 23–25.

POSTER PRESENTATIONS

- Garre-Frutos, F., Vadillo, M. A., Theeuwes, J., Van Moorselaar, D., & Lupiáñez, J. (2025). *Selective attention and explicit awareness independently contribute to the learning process of reward-related attentional biases*. 28th Annual Meeting of the Association for the Scientific Study of Consciousness, Heraklion, Greece, July 6–9.
- Rey-Sáez, R., Garre-Frutos, F., Franco-Martínez, A., Castillejo, I., & Vadillo, M. A. (2025). *Addressing methodological challenges in unconscious process research: A hierarchical modeling approach*. 28th Annual Meeting of the Association for the Scientific Study of Consciousness, Heraklion, Greece, July 6–9.
- Rey-Sáez, R., Garre-Frutos, F., Franco-Martínez, A., Castillejo, I., & Vadillo, M. A. (2025). *Hierarchical models in experimental psychology: A solution for measuring unconscious psychological processes*. 14th Reunión Científica sobre Atención, Madrid, Spain, April 3–4.
- Garre-Frutos, F. (2025). *The role of selective attention and explicit awareness in reward-related attentional biases*. Meetings of the Alhambra: The Cognitive Neuroscience of Consciousness, Granada, Spain, April 25–26.
- Garre-Frutos, F. (2025). *Modelling Pavlovian attentional biases through reinforcement learning*. Reinforcement Learning and Decision-Making Conference, Dublin, Ireland, June 11–14.
- Garre-Frutos, F., & Ariza, A. (2025). *Persistence of incentive salience: Devaluation of Pavlovian-to-instrumental transfer (PIT) and reversal of value-modulated attentional capture (VMAC)*. XXXV International Congress of the Spanish Society for Comparative Psychology, San Sebastián, Spain, October 1–3.
- Ariza, A., Garre-Frutos, F., & González, F. (2024). *Persistence of the value-modulated attentional capture (VMAC) effect using a reward-only variant task and specific extinction of the increase-signal in reward magnitude*. XXXIV International Conference of the Spanish Society for Comparative Psychology, Baeza, Spain, September 25–27.

- **Garre-Frutos, F.,** Vadillo, M. A., Botta, F., & Lupiáñez, J. (2024). *The role of informational value on value-modulated attentional capture*. 27th Annual Meeting of the Association for the Scientific Study of Consciousness, Tokyo, Japan, July 2–5.
- **Garre-Frutos, F.,** Lupiáñez, J., & Vadillo, M. A. (2024). *Is there value in value-driven distraction? A meta-analysis of studies on individual differences*. XIV SEPEX Conference, Almería, Spain, October 23–25.
- **Garre-Frutos, F.,** Lupiáñez, J., & Vadillo, M. A. (2024). *Value-modulated attentional capture depends on explicit awareness*. XIV SEPEX Conference, Almería, Spain, October 23–25.
- Ariza, A., Martínez-López, P., **Garre-Frutos, F.,** Luque, D., Vadillo, M. A., & González, F. (2024). *Explicit knowledge modulates reversal learning in value-modulated attentional capture*. XIV SEPEX Conference, Almería, Spain, October 23–25.
- Martínez-López, P., Vázquez-Millán, A., **Garre-Frutos, F.,** & Luque, D. (2024). *Using a forced-response task to measure habit formation in humans*. XIV SEPEX Conference, Almería, Spain, October 23–25.
- **Garre-Frutos, F.,** & Lupiáñez, J. (2023). *Inhibition of return and learned value*. 22nd Conference of the European Society for Cognitive Psychology, Porto, Portugal, August 24–27.
- Manini, G., **Garre-Frutos, F.,** Botta, F., Martín-Arévalo, E., & Lupiáñez, J. (2023). *Interference vs. distraction: Qualitatively different attentional capture effects depending on distractor relevance*. 22nd Conference of the European Society for Cognitive Psychology, Porto, Portugal, August 24–27.
- **Garre-Frutos, F.,** Hinojosa-Aguayo, I., González, F., Vadillo, M. A., & Lupiáñez, J. (2022). *A VMAC online replication and multiverse reliability analysis*. International APPE–SEPEX Meeting, Faro, Portugal, May 5–7.

Teaching

Attention and Memory

UNIVERSITAT OBERTA DE CATALUNYA

6 ECTS

2025 – 2026

Degree: BSc Psychology.

Behavioural Analysis and Motor Development in Physical Activity

UNIVERSITY OF GRANADA

4 ECTS

2024 – 2025

Degree: BSc in Physical Activity and Sport Sciences.

Perception and Attention

UNIVERSITY OF GRANADA

3 ECTS/year

2022/23 – 2023/24

Degree: BSc Psychology.

Student Supervision

María López Machuca

BSC STUDENT MENTORING

University of Granada

2026

Project: *On the Nature of Conscious Experience: A Comparison of Subjective Report Methods.*

Supervisors: Daniel Sanabria Lucena and Francisco Garre-Frutos.

Pablo Martínez-López

MSC THESIS SUPERVISION

University of Granada

2023

Project: *Within-Subjects Experimental Study of Individual Differences in Value-Modulated Attentional Capture (VMAC) and its Reversal.*

Supervisors: Felisa González and Francisco Garre-Frutos.

Adriana Ariza

MSC THESIS SUPERVISION

University of Granada

2023

Project: *Pavlovian-to-Instrumental Transfer and Implicit Emotion Regulation.*

Supervisors: Felisa González and Francisco Garre-Frutos.

Academic Events Organized

II CIMCYC Workshop on computational modelling of behavioral data

WORKSHOP · MIND, BRAIN, AND BEHAVIOR RESEARCH CENTER

Granada, Spain

2025

Measuring individual differences with experimental tasks: current pitfalls and future directions

Almería, Spain

SYMPOSIUM · XIV CONGRESO DE LA SOCIEDAD ESPAÑOLA DE PSICOLOGÍA EXPERIMENTAL

2024

Academic Service & Professional Memberships

Junior Editor, *Psicológica*

EDITORIAL SERVICE

Psicológica

Since 2025

Journal: *Psicológica*.

Role: Junior editorial role supporting the journal's peer-review and editorial workflow.

Ad hoc reviewer

PEER REVIEW

3 journals

Since 2025

Journals: *Journal of Experimental Psychology: Human Perception and Performance*, *Psychological Research*, and *Behavior Research Methods*.

Professional Societies

PROFESSIONAL MEMBERSHIPS

Memberships: Association for Psychological Science; Sociedad Española de Psicología Experimental; Association for the Scientific Study of Consciousness.

Outreach Activities

Psychobeers

PUBLIC ENGAGEMENT

Role: Organizing committee member for public engagement events connecting cognitive science and psychology with general audiences.

European Night of the Researchers

SCIENCE COMMUNICATION

Activity: Participant in *así funciona tu atención (y así te engaña)*, focused on communicating attention research to the public.